MoneyFollows

A life-simulation game for financial literacy

M2: Progress Report · Team Magadon

Problem statement: Vanguard India, *Empowering the Next Generation Through Financial Literacy*

1. Current approach and changes since M1

The core approach is unchanged: students run a persona's money week by week, see the consequences of each choice with a short lesson, and can rewind. The game engine is deterministic, content is data in English and Hindi, AI is used only at authoring time, and the product is delivered through NGOs. What changed:

- **Name:** the product is now *MoneyFollows*.
- **Backend:** we built a first version as our own API server, then moved to Supabase (managed PostgreSQL in Mumbai). Access rules now live in the database as row-level security, and the app calls a few SQL functions. This means less infrastructure to run and secure.
- **Game world:** the app opens on the character's room, with a Play button and a news rack. Play opens an illustrated map of the neighbourhood (home, shop, school, hospital; a bank is planned), and decisions happen inside these places.
- **In-game phone:** messages and calls mix scams with genuine offers, so students spot scams in context. A learning app on the phone holds the growth explorer, always shown as a low, average and high range with an assumption banner.
- **Paisa Times:** a weekly in-game newspaper of five short, original explainers. Tapping a highlighted word explains it, and answering each article's question pays the character's paper-route wage.
- **School classroom:** a practice quiz plus the start- and end-of-term checks. Separate question banks keep practice from inflating the measured learning gain.

Built so far.

- **Student app:** onboarding (language, class code, nickname, avatar, persona), room, map, phone, Paisa Times, School classroom, simple life meters, and an offline event queue with a sync banner.
- **Engine:** the full weekly loop (pins, decisions, delayed effects, rewind, sleep, month report, Money Health Score, growth ranges). It is tested but not yet wired into the map screens.
- **Dashboard:** staff sign-in, class creation with a code, QR and printable join sheet, a progress table, and learning gains by pillar.
- **Backend:** 7 tables, all under row-level security, and 10 SQL functions (join, sync, progress, learning gain and more).
- **Content:** 24 scenarios (6 per pillar), 10 life events, one newspaper edition (5 articles), 20 quiz questions and 12 scam messages, with all 569 strings in both English and Hindi.

2. Experiments run so far and results

- **Engine stress run:** 300 full-year games (50 seeds × 2 personas × 3 levels) on the real content. Every game finishes, balances never go negative, and every save reloads exactly.
- **Money conservation:** 120 games with random content and choices. At every step, money in equals money out.
- **AI-drafted content:** all 82 content items were drafted by an AI model as JSON and pass the content linter (schema, English and Hindi coverage, guardrail word lists). None has been human-reviewed yet; a review sheet is ready for reviewers.
- **Backend choice:** a custom API server worked but needed its own hosting and auth. Supabase gave us both, and the database itself enforces who sees which class.
- **App size:** the student app's code is 185 KB compressed, under our 250 KB budget.
- **Bug hunt:** we found and fixed a crash that blanked the app in newer Chrome; a lint rule now blocks the cause.
- **Test quality check:** we broke the classroom quiz logic in four ways; the tests caught all four.

3. Progress on testing

- **Overall:** 812 automated tests in 53 files, all passing. Lint and the content linter also pass.
- **Engine:** determinism, the money rules, rewind, delayed effects, a recorded 12-month golden game, and the stress and conservation runs above.
- **Backend:** 74 database tests run against Supabase's own PostgreSQL image. They cover joining a class, duplicate-free sync, facilitators seeing only their own classes, the nickname filter and the dashboard data.
- **Apps:** 30 test files for the student app, the dashboard and the sync client (offline queue, join and rejoin, classroom, phone, sign-in), plus a browser check of map → School → quiz in English and Hindi, including tap-target sizes.
- **Content:** 0 linter errors across 15 content files in both languages. The strict release check will pass only after human review.
- **Not yet done:** offline play on a real low-end phone, a first-time user test, the Hindi spot-check, and restoring our CI runs (paused by a GitHub account limit; all checks currently run locally).

4. Plan for the remaining time

- **Until feature freeze (08:30):**
  - Connect the engine's weekly loop to the map: pinned places, decision scenes, consequence and rewind, sleep at home, and the month recap with the Money Health Score.
  - Add the bank (growth explorer), the shop's sub-map and the Scam or Safe? mini-game (its content is ready).
  - Make the app load offline, add a seeded demo mode, and deploy a live link.
  - Human review of content and Hindi, then the strict content check.
- **After freeze (until 10:30):** real-phone and first-time-user tests, bug fixes only, LICENSE and README, the final report and the demo video.